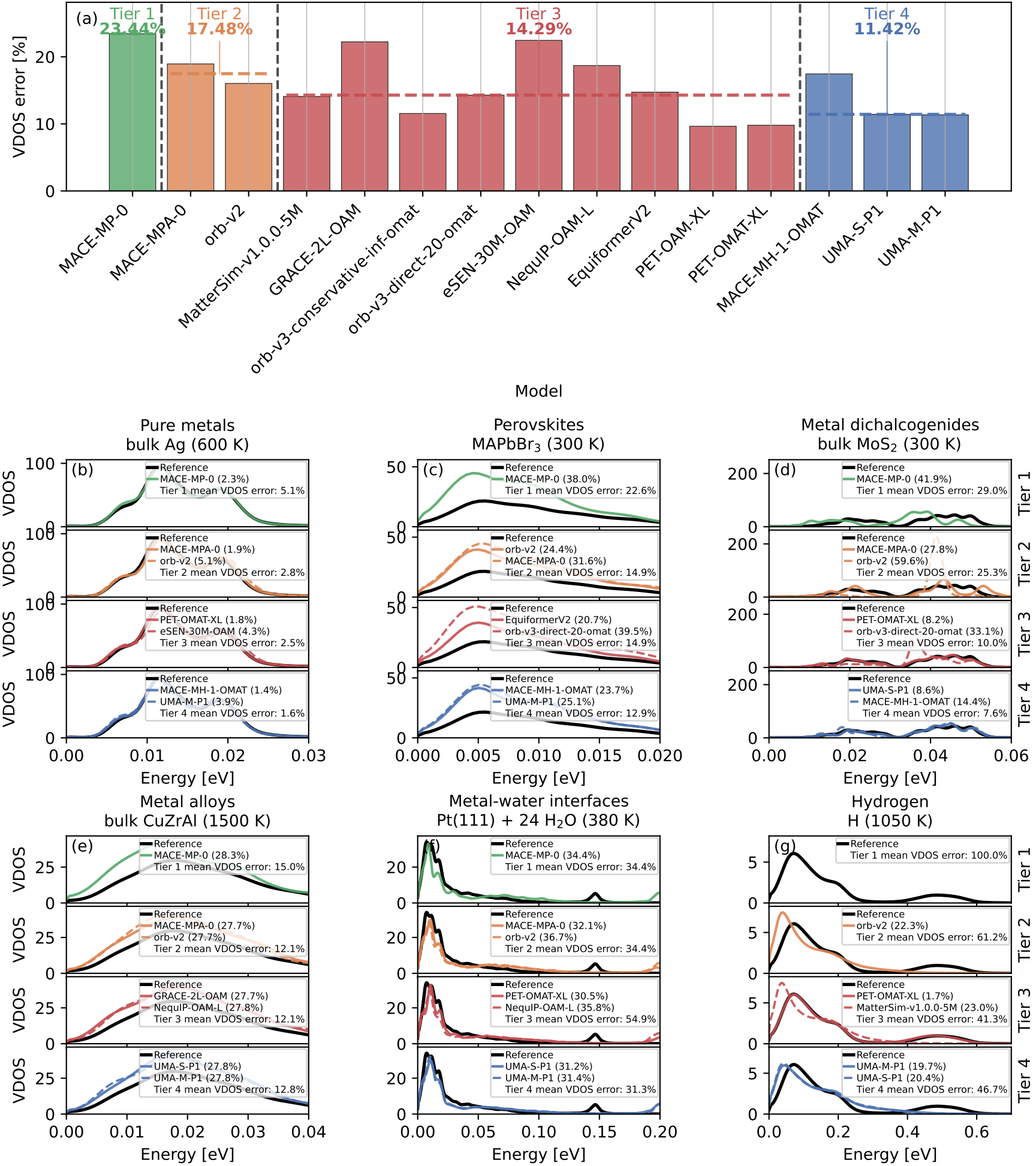

(a)
Tier 1
23.44%
Tier 2
17.48%
Tier 3
14.29%
Tier 4
11.42%
VDOS error [%]
MACE-MP-0
MACE-MPA-0
orb-v2
MatterSim-v1.0.0-5M
GRACE-2L-OAM
orb-v3-conservative-inf-omat
orb-v3-direct-20-omat
eSEN-30M-OAM
NequIP-OAM-L
EquiformerV2
PET-OAM-XL
PET-OMAT-XL
MACE-MH-1-OMAT
UMA-S-P1
UMA-M-P1
Model
Pure metals
bulk Ag (600 K)
(b)
Reference
MACE-MP-0 (2.3%)
Tier 1 mean VDOS error: 5.1%
MACE-MPA-0 (1.9%)
orb-v2 (5.1%)
Tier 2 mean VDOS error: 2.8%
PET-OMAT-XL (1.8%)
eSEN-30M-OAM (4.3%)
Tier 3 mean VDOS error: 2.5%
MACE-MH-1-OMAT (1.4%)
UMA-M-P1 (3.9%)
Tier 4 mean VDOS error: 1.6%
VDOS
Energy [eV]
Perovskites
MAPbBr3 (300 K)
(c)
MACE-MP-0 (38.0%)
Tier 1 mean VDOS error: 22.6%
orb-v2 (24.4%)
MACE-MPA-0 (31.6%)
Tier 2 mean VDOS error: 14.9%
EquiformerV2 (20.7%)
orb-v3-direct-20-omat (39.5%)
Tier 3 mean VDOS error: 14.9%
MACE-MH-1-OMAT (23.7%)
UMA-M-P1 (25.1%)
Tier 4 mean VDOS error: 12.9%
Metal dichalcogenides
bulk MoS2 (300 K)
(d)
MACE-MP-0 (41.9%)
Tier 1 mean VDOS error: 29.0%
MACE-MPA-0 (27.8%)
orb-v2 (59.6%)
Tier 2 mean VDOS error: 25.3%
PET-OMAT-XL (8.2%)
orb-v3-direct-20-omat (33.1%)
Tier 3 mean VDOS error: 10.0%
UMA-S-P1 (8.6%)
MACE-MH-1-OMAT (14.4%)
Tier 4 mean VDOS error: 7.6%
Tier 1
Tier 2
Tier 3
Tier 4
Metal alloys
bulk CuZrAl (1500 K)
(e)
MACE-MP-0 (28.3%)
Tier 1 mean VDOS error: 15.0%
MACE-MPA-0 (27.7%)
orb-v2 (27.7%)
Tier 2 mean VDOS error: 12.1%
GRACE-2L-OAM (27.7%)
NequIP-OAM-L (27.8%)
Tier 3 mean VDOS error: 12.1%
UMA-S-P1 (27.8%)
UMA-M-P1 (27.8%)
Tier 4 mean VDOS error: 12.8%
Metal-water interfaces
Pt(111) + 24 H2O (380 K)
(f)
MACE-MP-0 (34.4%)
Tier 1 mean VDOS error: 34.4%
MACE-MPA-0 (32.1%)
orb-v2 (36.7%)
Tier 2 mean VDOS error: 34.4%
PET-OMAT-XL (30.5%)
NequIP-OAM-L (35.8%)
Tier 3 mean VDOS error: 54.9%
UMA-S-P1 (31.2%)
UMA-M-P1 (31.4%)
Tier 4 mean VDOS error: 31.3%
Hydrogen
H (1050 K)
(g)
Tier 1 mean VDOS error: 100.0%
orb-v2 (22.3%)
Tier 2 mean VDOS error: 61.2%
PET-OMAT-XL (1.7%)
MatterSim-v1.0.0-5M (23.0%)
Tier 3 mean VDOS error: 41.3%
UMA-M-P1 (19.7%)
UMA-S-P1 (20.4%)
Tier 4 mean VDOS error: 46.7%